Resting-State fMRI Data Quality and Analysis in the WAND 7T Dataset

A Comparative Evaluation of Preprocessing Pipelines

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The temporal signal-to-noise ratio (tSNR) is defined voxel-wise as:

$$\text{tSNR}(\mathbf{x}) = \frac{\bar{S}(\mathbf{x})}{\sigma_t(\mathbf{x})} \tag{1}$$

where $\bar{S}(\mathbf{x})$ is the temporal mean and $\sigma_t(\mathbf{x})$ is the temporal standard deviation of the BOLD signal at voxel $\mathbf{x}$. The group-level metric reported is the median tSNR across all brain voxels.

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Table 1: tSNR for subject sub-01187 under four preprocessing conditions. MRIQC value serves as the reference standard.

Condition	**tSNR**	**Δ vs MRIQC**
Raw + simple intensity mask (baseline)	21.18	−7.43 (−26.0%)
Raw + nilearn EPI mask	25.78	−2.83 (−9.9%)
Moco + simple intensity mask	26.85	−1.77 (−6.2%)
Moco + nilearn EPI mask	31.45	+2.83 (+9.9%)
MRIQC reference	28.62	—

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